



6067 - Dynamics of high-redshift proto-globular clusters

Cycle: 3, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

The first JWST observations have already led to the detection of a large number of possible globular cluster (GC) progenitors at high redshifts. These observations are significantly enriching the observational study of GCs by providing key information on the fundamental properties of these systems and their external environment at the epoch of their formation.

The proposed investigation is aimed at building a comprehensive theoretical framework to determine the possible evolutionary links between the high-redshift proto-GCs observed by JWST and old GCs in the local universe. We will combine high-resolution cosmological simulations with realistic N-body and Monte Carlo simulations to carry out an extensive investigation of the dynamical evolution of GCs. In the first part we will run a large survey of simulations aimed at studying the early dynamics of GCs and explore when the strong external tidal field (as constrained by cosmological simulations) in the high-redshift environment where they form (and its time variation) may lead to the clusters' rapid dissolution. For surviving clusters, we will study how the high-redshift external tidal field affects the clusters' structural and kinematic properties emerging at the end of these early evolutionary phases. Our investigation will then continue with simulations following of the long-term evolution of the surviving clusters and provide all the elements necessary to determine the possible evolutionary paths linking the GC progenitors observed by JWST at high-redshifts and local old GCs.

OBSERVING DESCRIPTION